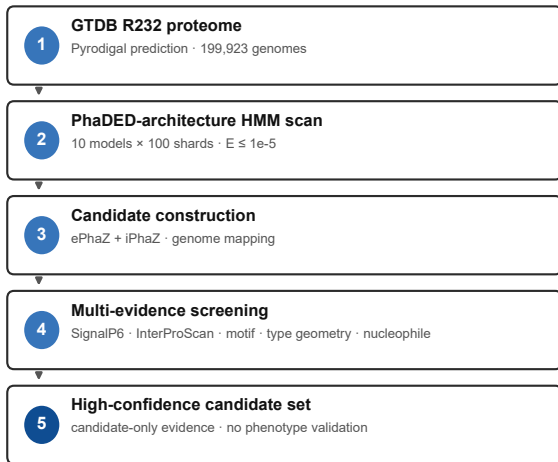


a*Process view***b****1 GTDB R232 proteome** $\approx 2.92 \times 10^8$

proteins · 199,923 genomes · 100 shards / 267 GB

2 PhaDED-architecture HMM scan

6,595,162

unique proteins · 6,743,198 hit rows · 10 models × 100 shards · E ≤ 1e-5

3 Candidate construction

109,087

candidate proteins · 62,666 genomes · ePhaZ + iPhaZ deduplicated

4 Multi-evidence screening

40,690

high-confidence proteins · 30,623 genomes

Screening detail: 36,611 pass + 71,860 hold + 616 unassigned = 109,087 (52 Cys demoted);
4,131 additional strong-tier hits screened under the same criteria.